

Time-Periodic Solutions of a KdV Equation at a Critical Length

Problem Statement

Consider the Korteweg–de Vries (KdV) equation on the interval $x \in (0, L)$:

$$y_t + y_{xxx} + y_x + y y_x = 0, \quad (1)$$

with boundary conditions

$$y(t, 0) = y(t, L) = y_x(t, L) = 0, \quad (2)$$

where L belongs to the critical length set

$$\mathcal{N} := \left\{ 2\pi \sqrt{\frac{k^2 + kl + l^2}{3}} : k, l \in \mathbb{Z}_{>0} \right\}$$

introduced by Rosier (1997).

Task. Take $k = 2, l = 1$, so that

$$L = 2\pi \sqrt{\frac{4 + 2 + 1}{3}} = 2\pi \sqrt{\frac{7}{3}}.$$

Prove the existence of a non-zero, time-periodic solution (y, T) with $T > 0$ to equations (??)–(??) with this value of L .

Overview of the proof. Rather than employing the Lyapunov–Schmidt bifurcation approach suggested by the hint, we observe that any nonzero stationary solution $y(t, x) = u(x)$ is automatically T -periodic for every $T > 0$. We reduce the PDE to a second-order ODE boundary value problem via a first integral, construct a one-parameter family of negative-loop orbits using phase-plane analysis, and show by the intermediate value theorem that one member of this family has spatial period exactly $L = 2\pi\sqrt{7/3}$.

1 Stationary Profile Criterion

Lemma 1. *Let $L > 0$ and suppose there exist $C \in \mathbb{R}$ and a nonzero function $u \in C^3([0, L])$ such that*

$$u''(x) + u(x) + \frac{1}{2} u(x)^2 = C, \quad u(0) = u'(0) = u(L) = u'(L) = 0.$$

Then for every $T > 0$, the function $y(t, x) = u(x)$ is a nonzero T -periodic classical solution of

$$y_t + y_{xxx} + y_x + y y_x = 0, \quad y(t, 0) = y(t, L) = y_x(t, L) = 0.$$

Proof. Let $T > 0$ be arbitrary and define

$$y(t, x) := u(x), \quad (t, x) \in \mathbb{R} \times [0, L].$$

Since $u \in C^3([0, L])$, the function y is classical in the sense needed for the PDE: it is C^1 in t and C^3 in x , with

$$y_t(t, x) = 0, \quad y_x(t, x) = u'(x), \quad y_{xxx}(t, x) = u'''(x).$$

Now differentiate the given integrated ODE

$$u''(x) + u(x) + \frac{1}{2} u(x)^2 = C$$

with respect to x . Since $u \in C^3([0, L])$ and C is constant, for $x \in (0, L)$ we get

$$u'''(x) + u'(x) + u(x) u'(x) = 0.$$

Therefore, for every $(t, x) \in \mathbb{R} \times (0, L)$,

$$y_t + y_{xxx} + y_x + y y_x = 0 + u'''(x) + u'(x) + u(x) u'(x) = 0.$$

The boundary conditions follow immediately from the hypotheses on u :

$$y(t, 0) = u(0) = 0, \quad y(t, L) = u(L) = 0, \quad y_x(t, L) = u'(L) = 0.$$

Finally, since y is independent of t , for every $T > 0$,

$$y(t + T, x) = u(x) = y(t, x),$$

so y is T -periodic in time. Because u is nonzero, the function $y(t, x) = u(x)$ is also nonzero. $\square$

2 Construction of Negative-Loop Stationary Profiles

Proposition 2. For every $a \in (0, 3/2)$, define

$$C_a = -\frac{a}{2} + \frac{a^2}{6}$$

and

$$P(a) = 2 \int_0^a \frac{dv}{\sqrt{v(a-v)(1 - \frac{a+v}{3})}}.$$

There exists a nonzero function $u_a \in C^3([0, P(a)])$ such that

$$u_a'' + u_a + \frac{1}{2} u_a^2 = C_a, \quad u_a(0) = u_a'(0) = u_a(P(a)) = u_a'(P(a)) = 0,$$

and $\min_{[0, P(a)]} u_a = -a$.

Proof. Fix $a \in (0, 3/2)$, and set $C = C_a = -\frac{a}{2} + \frac{a^2}{6}$. Define

$$G(v) = v(a-v) \left(1 - \frac{a+v}{3} \right), \quad 0 \leq v \leq a.$$

Phase-plane energy. The usual phase-plane energy for $u'' + u + \frac{1}{2}u^2 = C$ is

$$E(u, u') = \frac{1}{2}(u')^2 + \frac{1}{2}u^2 + \frac{1}{6}u^3 - Cu.$$

Indeed, for any C^2 solution,

$$\frac{d}{dx} E(u(x), u'(x)) = u'(x)(u''(x) + u(x) + \frac{1}{2}u(x)^2 - C) = 0.$$

Thus any solution satisfying $u = 0$, $u' = 0$ at an endpoint must lie on the zero-energy curve

$$(u')^2 = 2Cu - u^2 - \frac{1}{3}u^3.$$

Factoring the right-hand side. Using $C = -a/2 + a^2/6$,

$$2Cu - u^2 - \frac{1}{3}u^3 = -\frac{1}{3}u(u+a)(u+3-a).$$

For the negative branch, write $v = -u$. Then the zero-energy relation becomes

$$(v')^2 = G(v) = v(a-v) \left(1 - \frac{a+v}{3} \right).$$

Positivity of G . Since $0 < a < 3/2$, for $0 \leq v \leq a$,

$$1 - \frac{a+v}{3} \geq 1 - \frac{2a}{3} > 0.$$

Hence

$$G(v) > 0 \text{ for } 0 < v < a, \quad G(0) = G(a) = 0.$$

Integrability. Let $m = 1 - \frac{2a}{3} > 0$. Then $G(v) \geq m v(a-v)$ for $0 \leq v \leq a$. Therefore

$$\frac{1}{\sqrt{G(v)}} \leq \frac{1}{\sqrt{m} \sqrt{v(a-v)}}.$$

The comparison function is integrable on $(0, a)$, since

$$\int_0^a \frac{dv}{\sqrt{v(a-v)}} = \pi.$$

Thus

$$T := \int_0^a \frac{dv}{\sqrt{G(v)}} < \infty,$$

and by the definition of $P(a)$, $P(a) = 2T$.

Construction of the ascending branch. Define, for $0 \leq r \leq a$,

$$I(r) = \int_0^r \frac{dv}{\sqrt{G(v)}}.$$

This is understood as an improper integral at $v = 0$, and also at $v = a$ when $r = a$.

The function I is finite and continuous on $[0, a]$. Indeed, the integrand is dominated by the integrable function $(m v(a - v))^{-1/2}$. More explicitly, if $r_n \rightarrow r$, then

$$|I(r_n) - I(r)| \leq \int_{\min(r_n, r)}^{\max(r_n, r)} \frac{dv}{\sqrt{G(v)}},$$

and the right-hand side tends to 0 because the integrand is locally bounded in $(0, a)$ and has integrable endpoint singularities.

Moreover, I is strictly increasing: if $0 \leq r < s \leq a$, then

$$I(s) - I(r) = \int_r^s \frac{dv}{\sqrt{G(v)}} > 0,$$

because the integrand is positive on $(0, a)$. Hence I maps $[0, a]$ continuously and strictly increasingly onto $[0, T]$. Therefore it has a continuous inverse

$$w : [0, T] \rightarrow [0, a].$$

Thus $w(0) = 0$, $w(T) = a$, and $0 < w(t) < a$ for $0 < t < T$.

Differentiability and ODE. For $r \in (0, a)$, the integrand $G(r)^{-1/2}$ is smooth, so I is differentiable there and

$$I'(r) = \frac{1}{\sqrt{G(r)}} > 0.$$

Let $t \in (0, T)$. Since $w(t) \in (0, a)$, the inverse relation $I(w(t)) = t$ may be differentiated. Using the mean value theorem, for small h ,

$$h = I(w(t+h)) - I(w(t)) = I'(\xi_h)(w(t+h) - w(t))$$

for some ξ_h between $w(t+h)$ and $w(t)$. As $h \rightarrow 0$, $\xi_h \rightarrow w(t)$, so

$$w'(t) = \frac{1}{I'(w(t))} = \sqrt{G(w(t))}.$$

Therefore $w'(t) > 0$ for $0 < t < T$.

Since G is a polynomial and $G > 0$ on $(0, a)$, the map $v \mapsto \sqrt{G(v)}$ is smooth on $(0, a)$. Hence $w \in C^\infty((0, T))$. Differentiating $w' = \sqrt{G(w)}$ gives

$$w'' = \frac{G'(w)}{2\sqrt{G(w)}} w' = \frac{1}{2} G'(w).$$

Now

$$G(v) = \left(a - \frac{a^2}{3}\right)v - v^2 + \frac{1}{3}v^3,$$

so $G'(v) = a - \frac{a^2}{3} - 2v + v^2$. Differentiating once more,

$$w''' = \frac{1}{2} G''(w) w'.$$

Endpoint regularity. Since $w(t) \rightarrow 0$ as $t \downarrow 0$ and $w(t) \rightarrow a$ as $t \uparrow T$, and since $w'(t) = \sqrt{G(w(t))}$, we get

$$\lim_{t \downarrow 0} w'(t) = 0, \quad \lim_{t \uparrow T} w'(t) = 0.$$

Also,

$$\lim_{t \downarrow 0} w''(t) = \frac{1}{2} G'(0) = \frac{a}{2} - \frac{a^2}{6},$$

and

$$\lim_{t \uparrow T} w''(t) = \frac{1}{2} G'(a) = -\frac{a}{2} + \frac{a^2}{3}.$$

Finally, because G'' is bounded on $[0, a]$ and $w'(t) \rightarrow 0$ at both endpoints,

$$\lim_{t \downarrow 0} w'''(t) = 0, \quad \lim_{t \uparrow T} w'''(t) = 0.$$

The mean value theorem now identifies these one-sided limits with the actual endpoint derivatives. For example, since w' has a finite right-limit at 0,

$$\frac{w(h) - w(0)}{h} = w'(\xi_h) \rightarrow 0$$

for some $\xi_h \in (0, h)$. Hence $w'(0) = 0$. Applying the same argument successively to w' and w'' , and similarly at T from the left, gives

$$w \in C^3([0, T]),$$

with $w'(0) = w'(T) = 0$, $w'''(0) = w'''(T) = 0$.

The half-profile u_1 . Define $u_1(t) = -w(t)$ for $0 \leq t \leq T$. Then $u_1 \in C^3([0, T])$, and

$$u_1(0) = 0, \quad u_1'(0) = 0, \quad u_1(T) = -a, \quad u_1'(T) = 0.$$

For $0 < t < T$,

$$u_1''(t) = -w''(t) = -\frac{1}{2} G'(w(t)).$$

Using the formula for G' ,

$$-\frac{1}{2} G'(w) = -\frac{a}{2} + \frac{a^2}{6} + w - \frac{1}{2} w^2 = C + w - \frac{1}{2} w^2.$$

Since $u_1 = -w$, this becomes $u_1'' = C - u_1 - \frac{1}{2}u_1^2$. Therefore

$$u_1'' + u_1 + \frac{1}{2}u_1^2 = C$$

on $(0, T)$. Since $u_1 \in C^2([0, T])$, the left-hand side is continuous, so the same identity holds on all of $[0, T]$.

Reflection and the full profile u_a . Reflect u_1 evenly about $t = T$. Define

$$u_a(t) = \begin{cases} u_1(t), & 0 \leq t \leq T, \\ u_1(2T - t), & T \leq t \leq 2T. \end{cases}$$

For $t > T$, putting $s = 2T - t$, we have

$$u_a'(t) = -u_1'(s), \quad u_a''(t) = u_1''(s), \quad u_a'''(t) = -u_1'''(s).$$

At $t = T$, the one-sided derivatives agree through order 3: the zeroth and second derivatives agree automatically, while $u_1'(T) = 0$ and $u_1'''(T) = 0$. Thus $u_a \in C^3([0, 2T])$.

On $[0, T]$, we already know $u_a'' + u_a + \frac{1}{2}u_a^2 = C$. On $(T, 2T)$, using $s = 2T - t$,

$$u_a''(t) + u_a(t) + \frac{1}{2}u_a(t)^2 = u_1''(s) + u_1(s) + \frac{1}{2}u_1(s)^2 = C.$$

Since $u_a \in C^2([0, 2T])$, the identity also holds at the gluing point $t = T$ and at $t = 2T$.

Because $2T = P(a)$, we have constructed $u_a \in C^3([0, P(a)])$ satisfying $u_a'' + u_a + \frac{1}{2}u_a^2 = C_a$. The boundary conditions follow:

$$u_a(0) = u_1(0) = 0, \quad u_a'(0) = u_1'(0) = 0,$$

and

$$u_a(P(a)) = u_a(2T) = u_1(0) = 0, \quad u_a'(P(a)) = -u_1'(0) = 0.$$

Finally, w is strictly increasing from 0 to a on $[0, T]$. Hence $u_1 = -w$ decreases from 0 to $-a$, and the reflected half has the same range. Therefore

$$\min_{[0, P(a)]} u_a = -a.$$

Since $a > 0$, the function u_a is nonzero. □

3 Matching the Critical Length

Proposition 3. *The function*

$$P(a) = 2 \int_0^a \frac{dv}{\sqrt{v(a-v)(1 - \frac{a+v}{3})}} \quad (0 < a < 3/2)$$

is continuous, satisfies

$$\lim_{a \rightarrow 0^+} P(a) = 2\pi, \quad \lim_{a \rightarrow (3/2)^-} P(a) = +\infty,$$

and therefore there exists $a_* \in (0, 3/2)$ such that

$$P(a_*) = 2\pi\sqrt{\frac{7}{3}}.$$

Proof. Step 1: The basic endpoint integral

Let

$$g(s) = \frac{1}{\sqrt{s(1-s)}}, \quad 0 < s < 1.$$

Define $\mathcal{G}(s) = 2 \arcsin \sqrt{s}$. For $0 < s < 1$,

$$\mathcal{G}'(s) = 2 \cdot \frac{1}{\sqrt{1-s}} \cdot \frac{1}{2\sqrt{s}} = \frac{1}{\sqrt{s(1-s)}} = g(s).$$

Thus, for every $0 < r < t < 1$,

$$\int_r^t g(s) ds = \mathcal{G}(t) - \mathcal{G}(r).$$

Since $\lim_{s \rightarrow 0^+} \mathcal{G}(s) = 0$ and $\lim_{s \rightarrow 1^-} \mathcal{G}(s) = \pi$, the improper integral exists and

$$\int_0^1 \frac{ds}{\sqrt{s(1-s)}} = \pi.$$

Moreover, the endpoint tails vanish: for $0 < \eta < 1$,

$$\int_0^\eta g(s) ds = 2 \arcsin \sqrt{\eta} \rightarrow 0 \quad (\eta \rightarrow 0^+),$$

and

$$\int_{1-\eta}^1 g(s) ds = \pi - 2 \arcsin \sqrt{1-\eta} \rightarrow 0 \quad (\eta \rightarrow 0^+).$$

Step 2: Fixed-interval representation of $P(a)$

For $0 < a < 3/2$, define

$$F_a(s) = \frac{1}{\sqrt{s(1-s)\left(1 - \frac{a(1+s)}{3}\right)}}, \quad 0 < s < 1.$$

For $0 < s < 1$, $1 - \frac{a(1+s)}{3} \geq 1 - \frac{2a}{3} > 0$, so F_a is well-defined and positive. Also,

$$F_a(s) \leq \frac{1}{\sqrt{1 - \frac{2a}{3}}} g(s).$$

Since g has a finite improper integral on $(0, 1)$, the comparison test shows that $\int_0^1 F_a(s) ds$ exists as an improper integral.

Now justify the substitution $v = as$. On every compact subinterval of $(0, a)$, this is an ordinary C^1 change of variables. If $0 < \alpha < a/2$, then

$$\int_{\alpha}^{a/2} f_a(v) dv = \int_{\alpha/a}^{1/2} F_a(s) ds,$$

where $f_a(v) = (v(a-v)(1 - \frac{a+v}{3}))^{-1/2}$. Letting $\alpha \rightarrow 0^+$ is justified because

$$0 \leq \int_0^{\alpha/a} F_a(s) ds \leq \frac{1}{\sqrt{1 - \frac{2a}{3}}} \int_0^{\alpha/a} g(s) ds \rightarrow 0.$$

Similarly, the upper endpoint gives $\int_{a/2}^a f_a(v) dv = \int_{1/2}^1 F_a(s) ds$. Combining:

$$P(a) = 2 \int_0^1 \frac{ds}{\sqrt{s(1-s)\left(1 - \frac{a(1+s)}{3}\right)}}.$$

Step 3: Continuity of P

Write $F_a(s) = g(s) Q(a, s)$, where $Q(a, s) = \left(1 - \frac{a(1+s)}{3}\right)^{-1/2}$. Fix $a_0 \in (0, 3/2)$. Choose β such that $a_0 < \beta < \frac{3}{2}$ and set $c = 1 - \frac{2\beta}{3} > 0$. For $0 \leq x \leq \beta$ and $0 < s < 1$, $1 - \frac{x(1+s)}{3} \geq c$. Moreover,

$$\frac{\partial Q}{\partial x}(x, s) = \frac{1+s}{6} \left(1 - \frac{x(1+s)}{3}\right)^{-3/2},$$

so $\left|\frac{\partial Q}{\partial x}(x, s)\right| \leq \frac{1}{3} c^{-3/2} =: \Lambda$. By the mean value theorem, for $0 < a < \beta$,

$$|F_a(s) - F_{a_0}(s)| \leq \Lambda |a - a_0| g(s).$$

Integrating and using $\int_0^1 g(s) ds = \pi$:

$$\left| \int_0^1 F_a(s) ds - \int_0^1 F_{a_0}(s) ds \right| \leq \Lambda \pi |a - a_0|.$$

Since $P(a) = 2 \int_0^1 F_a(s) ds$, the function P is continuous at a_0 . Because $a_0 \in (0, 3/2)$ was arbitrary, P is continuous on $(0, 3/2)$.

Step 4: The limit as $a \rightarrow 0^+$

For $0 < a < 3/2$ and $0 < s < 1$,

$$1 \leq \left(1 - \frac{a(1+s)}{3}\right)^{-1/2} \leq \frac{1}{\sqrt{1 - \frac{2a}{3}}}.$$

Hence $g(s) \leq F_a(s) \leq \frac{1}{\sqrt{1-2a/3}} g(s)$. Integrating:

$$\pi \leq \int_0^1 F_a(s) ds \leq \frac{\pi}{\sqrt{1-\frac{2a}{3}}}.$$

As $a \rightarrow 0^+$, the right-hand side tends to π . By the squeeze theorem,

$$\lim_{a \rightarrow 0^+} P(a) = 2\pi.$$

Step 5: The limit as $a \rightarrow (3/2)^-$

Let $\delta = \frac{3}{2} - a > 0$. For $0 < s < 1$,

$$1 - \frac{a(1+s)}{3} = \frac{1-s}{2} + \frac{\delta(1+s)}{3}.$$

Assume $0 < \delta < 1/4$. For $s \in [1/2, 1-2\delta]$ (which is nonempty), $1-s \geq 2\delta$ and $\frac{\delta(1+s)}{3} \leq \frac{1-s}{3}$, so

$$1 - \frac{a(1+s)}{3} \leq \frac{5}{6}(1-s).$$

Hence $s(1-s)(1 - \frac{a(1+s)}{3}) \leq \frac{5}{6}(1-s)^2$, giving

$$F_a(s) \geq \sqrt{\frac{6}{5}} \cdot \frac{1}{1-s}.$$

Thus

$$P(a) \geq 2\sqrt{\frac{6}{5}} \int_{1/2}^{1-2\delta} \frac{ds}{1-s} = 2\sqrt{\frac{6}{5}} \log\left(\frac{1}{4\delta}\right).$$

As $\delta \rightarrow 0^+$, $\log(\frac{1}{4\delta}) \rightarrow +\infty$, so $\lim_{a \rightarrow (3/2)^-} P(a) = +\infty$.

Step 6: Application of the intermediate value theorem

Let $\mathcal{T} = 2\pi\sqrt{\frac{7}{3}}$. Since $\sqrt{\frac{7}{3}} > 1$, we have $\mathcal{T} > 2\pi$.

From $\lim_{a \rightarrow 0^+} P(a) = 2\pi$, there exists $a_1 \in (0, 3/2)$ such that $P(a_1) < \mathcal{T}$.

From $\lim_{a \rightarrow (3/2)^-} P(a) = +\infty$, there exists $a_2 \in (a_1, 3/2)$ such that $P(a_2) > \mathcal{T}$.

The function P is continuous on $[a_1, a_2]$. Since $P(a_1) < \mathcal{T} < P(a_2)$, the intermediate value theorem gives some $a_* \in (a_1, a_2) \subset (0, 3/2)$ such that

$$P(a_*) = 2\pi\sqrt{\frac{7}{3}}.$$

□

4 Conclusion: Existence of a Time-Periodic Solution

Theorem 4. *There exists a non-zero, time-periodic solution (y, T) with $T > 0$ of the KdV equation*

$$y_t + y_{xxx} + y_x + y y_x = 0, \quad y(t, 0) = y(t, L) = y_x(t, L) = 0,$$

with $L = 2\pi\sqrt{7/3}$.

Proof. By Proposition ??, there exists $a_* \in (0, 3/2)$ such that $P(a_*) = 2\pi\sqrt{7/3} = L$.

By Proposition ??, there exists a nonzero function $u_{a_*} \in C^3([0, L])$ satisfying

$$u_{a_*}'' + u_{a_*} + \frac{1}{2} u_{a_*}^2 = C_{a_*}, \quad u_{a_*}(0) = u_{a_*}'(0) = u_{a_*}(L) = u_{a_*}'(L) = 0.$$

By Lemma ??, for any $T > 0$ (say $T = 1$), the function $y(t, x) = u_{a_*}(x)$ is a nonzero T -periodic classical solution of the KdV equation with the prescribed boundary conditions. $\square$

References

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